DO NOW — RECALL

Database and SQL retrieval challenge

Name: _____ Class: _____ Date: _____

1. What is a primary key?

2. What does INNER JOIN return?

3. What is a foreign key?

4. A 3×2 zero-indexed array: specify row index range, column range and cell count.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Uniquely identifies a row.

2. Model answer

Matched records according to join condition.

3. Model answer

Column referencing a key in a related table.

4. Model answer

Rows 0–2, columns 0–1, total six cells.

Vocabulary — know these terms

primary key — Unique row identifier.

foreign key — Related table key reference.

INNER JOIN — Returns matching joined rows.

WHERE — Filters query results.

NEW LEARNING

Read, annotate and explain

Understand

Use a primary key for each Student row and a matching foreign key in Score.

Trace

INNER JOIN produces one result for each matching StudentID; Cleo has no matching Score and is absent.

Evaluate

A WHERE clause can filter matched records; forgetting the join condition can produce unwanted combinations.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Match Tutor and Class records on TutorID.

Predict first

Tutor 1 matches two class rows

Model and check

Tutor 2 has no class and is excluded from INNER JOIN.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Define primary key, foreign key, INNER JOIN and WHERE. [4]

Question 2

Write an INNER JOIN for Student and Score above selecting Name and Mark; give output. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Explain why omitting a join condition may produce misleading results. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

INNER JOIN automatically includes rows without a key match.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.